



PRA: Quantitative Project Risk Analysis in R

Paul Govan 
Independent Researcher

Abstract

The **PRA** package for R provides a unified, open-source implementation of the core analytical methods used in quantitative project risk analysis: the Second Moment Method (SMM), Monte Carlo simulation (MCS), earned value management (EVM), contingency and sensitivity analysis, Bayesian risk inference and updating, sigmoidal learning curves, and design structure matrices (DSM). It operationalizes causal-network and structural-network frameworks for project risk in a dependency-light R package available on CRAN. Version 0.5.0 further adds an experimental probabilistic network module for forward simulation and Bayesian conditioning along causal chains, and exposes every analytical function as a tool over the Model Context Protocol (MCP), so that large-language-model agents can run project risk analyses through a documented set of skills. The package adheres to rOpenSci Statistical Software Review standards and is supported by a continuous-integration test suite running across macOS, Windows, and Ubuntu.

Keywords: project risk analysis, Monte Carlo simulation, earned value management, Bayesian inference, design structure matrix, R.

1. Introduction

Quantitative risk analysis is a foundational discipline in project management, enabling organizations to assess schedule and cost uncertainty, identify high-impact risk drivers, and set evidence-based contingency reserves. Govan (2014) and Govan and Damnjanovic (2016) introduced the first implementation of causal networks for project risk analysis grounded in the resource-based view, demonstrating that organizational capabilities and shared resources—not just schedule uncertainty—are the structural pathways through which risks propagate across project tasks. Extending this work, Govan and Damnjanovic (2020) developed structural network measures that quantify the topology of project risk networks, enabling systematic identification of high-centrality risk nodes.

Despite broad industry adoption of quantitative methods, open-source tooling for integrated

project risk analysis in R remains fragmented. Proprietary platforms such as Oracle Primavera Risk Analysis and Palisade @RISK provide integrated workflows but impose licensing costs and limit reproducibility. Within R, no single package unifies Monte Carlo schedule risk, earned value management, Bayesian risk inference, sigmoidal learning curves, and dependency structure analysis under a common API. The **mc2d** package (Pouillot and Delignette-Muller 2010) provides general-purpose two-dimensional Monte Carlo simulation but has no project-specific constructs. The **sensitivity** package (Iooss, Da Veiga, Janon, and Pujol 2024) implements variance-based indices for general functions but does not integrate with schedule or cost models. EVM metrics are absent from CRAN entirely. Bayesian packages such as **rstan** (Stan Development Team 2024) and **brms** (Bürkner 2017) operate at a high level of generality and require substantial expertise to apply in project risk contexts.

PRA addresses this gap. It provides a coherent, CRAN-validated framework of ten analytical modules covering the full quantitative project risk workflow—from uncertainty characterization and simulation to performance monitoring and Bayesian updating—while operationalizing the causal and network-theoretic frameworks of Govan (2014) and Govan and Damjanovic (2016, 2020) computationally. The package targets project analysts who work in R and require reproducible, auditable outputs compatible with standard reporting pipelines. It adheres to rOpenSci Statistical Software Review (SRR) standards (rOpenSci Statistical Software Peer Review 2021) and uses minimal hard dependencies: **mc2d** (Pouillot and Delignette-Muller 2010) for triangular distribution families and **minpack.lm** (Elzhov, Mullen, Spiess, and Bolker 2016) for nonlinear least squares fitting. The optional Model Context Protocol (MCP) server activates through **Suggests**-only dependencies.

The following table summarizes how **PRA** compares with the most closely related R packages and the dominant commercial platform.

Table: Comparison of **PRA** with related R packages and commercial tools. “Exp.” = experimental feature; “—” = not supported.

Feature	PRA	mc2d	sensitivity	rstan/brms	@RISK	:————— :— :— :— :— :— :— :— :— :—																																															
MCS schedule risk	Yes	Partial	—	—	Yes	EVM (11 metrics)	Yes	—	—	—	Yes	Bayesian updating	Yes	—	—	Yes	—	Sigmoidal learning curves	Yes	—	—	—	—	Design structure matrices	Yes	—	—	—	—	Probabilistic networks	Exp.	—	—	Yes	—	MCP tool server	Yes	—	—	—	—	Open-source (CRAN)	Yes	Yes	Yes	Yes	—	SRR-compliant	Yes	—	—	—	—

The package is installed from CRAN in the standard way:

```
R> install.packages("PRA")
R> library("PRA")
```

Optional dependencies—**ellmer** (Wickham *et al.* 2025), **mcptools**, and **jsonlite**—are only required for the MCP server, `pra_mcp_server()`.

The remainder of this article describes each analytical module in turn, with working code examples drawn from the same R session. Section~2 covers schedule and cost uncertainty methods (SMM, MCS, contingency, sensitivity). Section~3 describes earned value management. Section~4 covers Bayesian risk inference and updating. Section~5 describes sigmoidal learning curves. Section~6 covers design structure matrices. Section~7 introduces the experimental probabilistic network module. Section~8 describes the Model Context Protocol integration. Section~9 presents an integrated case study. Section~10 concludes.

2. Schedule and Cost Uncertainty

Two complementary methods are provided for quantifying schedule and cost uncertainty: the Second Moment Method for rapid analytical estimates, and Monte Carlo simulation for full distributional analysis. Contingency extraction and sensitivity analysis complete the workflow.

We illustrate the workflow with `building_project`, a bundled example dataset describing a mid-rise commercial building with six work packages, adapted from the construction-project examples in [Damnjanovic and Reinschmidt \(2020\)](#).

```
R> data("building_project", package = "PRA")
R> bp <- building_project
R> bp$task_names

[1] "Site Preparation & Earthworks" "Foundations"
[3] "Structural Frame"              "Building Envelope"
[5] "Mechanical/Electrical/Plumbing" "Interior Fit-out & Finishes"
```

2.1. Second Moment Method

The Second Moment Method (SMM) propagates task-level means and variances to project-level estimates using the Central Limit Theorem ([Benjamin and Cornell 2000](#)). For a project with n tasks:

$$E[X] = \sum_{i=1}^n E[X_i], \quad \text{Var}(X) = \sum_{i=1}^n \text{Var}(X_i) + 2 \sum_{i < j} \rho_{ij} \sigma_i \sigma_j$$

The six task durations are triangular, so we obtain their means and variances in closed form and propagate them with `smm()`:

```
R> tri_mean <- function(d) (d$a + d$b + d$c) / 3
R> tri_var <- function(d) (d$a^2 + d$b^2 + d$c^2 -
+                          d$a * d$b - d$a * d$c - d$b * d$c) / 18
R> task_means <- vapply(bp$task_distributions, tri_mean, numeric(1))
R> task_vars <- vapply(bp$task_distributions, tri_var, numeric(1))
R>
R> result <- smm(task_means, task_vars) # independent tasks
R> cat("Total mean:", round(result$total_mean, 2), "weeks\n")
```

Total mean: 71.67 weeks

```
R> cat("Total SD: ", round(result$total_std, 2), "weeks\n")
```

Total SD: 4.53 weeks

Positive correlation between tasks inflates the total variance; `smm()` accepts a correlation matrix and adds the covariance terms analytically:

```
R> result_cor <- smm(task_means, task_vars, bp$cor_mat)
R> cat("Total SD (correlated):", round(result_cor$total_std, 2), "weeks\n")
```

```
Total SD (correlated): 6.59 weeks
```

SMM is useful when only first and second moments are available and a rapid estimate is needed. When task distributions are non-normal or accurate tail behavior is required, Monte Carlo simulation is preferred.

2.2. Monte Carlo Simulation

`mcs()` propagates uncertainty through a project network by drawing samples from user-specified task distributions—triangular, normal, or uniform—over a large number of iterations (Vose 2008). By default the tasks are sampled independently; an optional correlation matrix is supported, though its current handling is approximate (Section~10).

```
R> set.seed(42)
R> results <- mcs(10000, bp$task_distributions)
R> cat("Simulated mean:", round(results$total_mean, 2), "weeks\n")
```

```
Simulated mean: 71.72 weeks
```

```
R> cat("Simulated SD:  ", round(results$total_sd, 2), "weeks\n")
```

```
Simulated SD: 4.57 weeks
```

The `mcs()` function returns an S3 object of class "mcs" with a `print()` method that summarizes percentiles. The full simulated distribution is available in `results$total_distribution`. On a modern laptop, 10,000 simulations of a six-task network complete in under one second; increasing the count to 100,000 provides more precise tail estimates at roughly ten-fold additional runtime. The default of 10,000 balances accuracy and speed for most practical analyses.

Numerical Validation

For independent tasks, the Central Limit Theorem makes the SMM estimates the exact moments of the total, so the Monte Carlo simulation should reproduce them. The two agree to within Monte Carlo error, validating the sampling engine:

```
R> cat(sprintf("Mean: SMM %.2f vs MCS %.2f\n",
+             result$total_mean, results$total_mean))
```

```
Mean: SMM 71.67 vs MCS 71.72
```

```
R> cat(sprintf("SD: SMM %.2f vs MCS %.2f\n",
+             result$total_std, results$total_sd))
```

```
SD: SMM 4.53 vs MCS 4.57
```

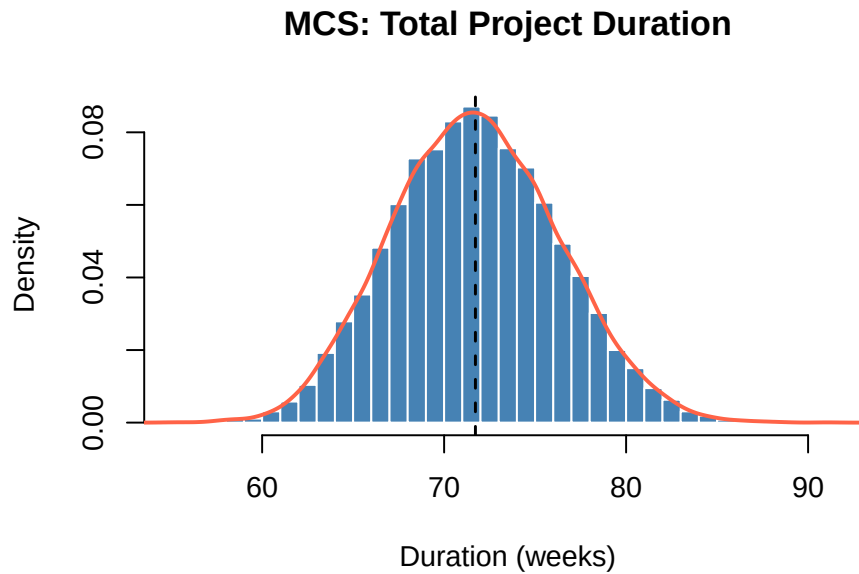


Figure 1: Distribution of simulated total project durations. The dashed vertical line marks the mean.

```
R> hist(results$total_distribution,
+   breaks = 50, freq = FALSE,
+   main = "MCS: Total Project Duration",
+   xlab = "Duration (weeks)", col = "steelblue", border = "white"
+ )
R> lines(density(results$total_distribution), col = "tomato", lwd = 2)
R> abline(v = results$total_mean, lty = 2, lwd = 1.5)
```

2.3. Contingency Analysis

`contingency()` extracts a reserve buffer as the difference between a high-confidence percentile and the base (median) estimate—a standard approach in project risk practice ([Project Management Institute 2021](#)).

```
R> reserve <- contingency(results, phigh = 0.80, pbase = 0.50)
R> cat("Schedule contingency (P80 - P50):", round(reserve, 2), "weeks\n")
```

```
Schedule contingency (P80 - P50): 3.98 weeks
```

2.4. Sensitivity Analysis

`sensitivity()` decomposes total project variance by task contribution using rank correlation, producing the inputs for a Tornado chart.

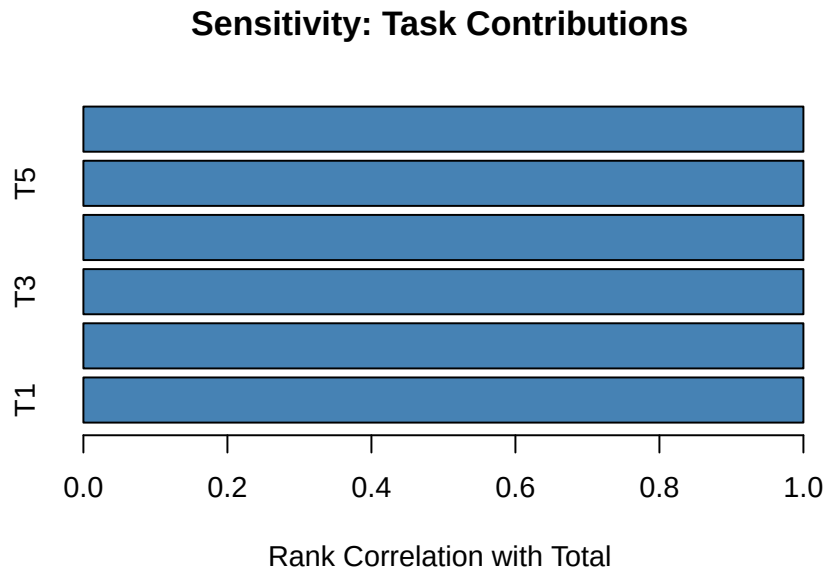


Figure 2: Tornado chart: each bar shows a task's rank correlation with total project duration.

```
R> sens <- sensitivity(bp$task_distributions)
R> names(sens) <- paste0("T", seq_along(sens))
R> barplot(sort(sens),
+   horiz = TRUE, col = "steelblue",
+   main = "Sensitivity: Task Contributions",
+   xlab = "Rank Correlation with Total"
+ )
```

Tasks with larger bars are the dominant drivers of total risk; mitigation resources should be directed there first.

3. Earned Value Management

The EVM module implements the complete ANSI/EIA-748 suite of eleven performance metrics (Fleming and Koppelman 2010). The three core quantities—Planned Value (PV), Earned Value (EV), and Actual Cost (AC)—are computed first; all performance indices and forecasts follow from them.

```
R> bac <- 500000 # budget at completion ($)
R> schedule <- c(0.10, 0.25, 0.50, 0.75, 1.00) # cumulative value curve
R> time_period <- 3

R> pv_val <- pv(bac, schedule, time_period)
R> ev_val <- ev(bac, actual_per_complete = 0.40)
R> ac_val <- ac(c(45000, 110000, 135000), time_period, cumulative = FALSE)
```

```

R>
R> cat("PV: $", format(pv_val, big.mark = ","), "\n")

PV: $ 250,000

R> cat("EV: $", format(ev_val, big.mark = ","), "\n")

EV: $ 2e+05

R> cat("AC: $", format(ac_val, big.mark = ","), "\n")

AC: $ 290,000

R> spi_val <- spi(ev_val, pv_val)
R> cpi_val <- cpi(ev_val, ac_val)
R> sv_val <- sv(ev_val, pv_val)
R> cv_val <- cv(ev_val, ac_val)
R>
R> cat("SPI:", round(spi_val, 3), " CPI:", round(cpi_val, 3), "\n")

SPI: 0.8 CPI: 0.69

R> cat("SV: $", format(round(sv_val), big.mark = ","),
+      " CV: $", format(round(cv_val), big.mark = ","), "\n")

SV: $ -50,000 CV: $ -90,000

Both  $SPI < 1$  and  $CPI < 1$  indicate the project is behind schedule and over budget. The
eac() function provides three completion forecasts under different efficiency assumptions:

R> cat("EAC (typical): $", format(round(eac(bac, method = "typical",
+    cpi = cpi_val))), big.mark = ","), "\n")

EAC (typical): $ 725,000

R> cat("EAC (atypical): $", format(round(eac(bac, method = "atypical",
+    ac = ac_val, ev = ev_val))), big.mark = ","), "\n")

EAC (atypical): $ 590,000

R> cat("EAC (combined): $", format(round(eac(bac, method = "combined",
+    cpi = cpi_val, ac = ac_val, ev = ev_val, spi = spi_val))),
+    big.mark = ","), "\n")

EAC (combined): $ 833,750

```

The remaining EVM metrics are:

```
R> eac_typ <- eac(bac, method = "typical", cpi = cpi_val)
R> cat("ETC:  $", format(round(etc(bac, ev_val, cpi = cpi_val)), big.mark = ","), "\n")
```

ETC: \$ 435,000

```
R> cat("TCPI:", round(tcpi(bac, ev_val, ac_val), 3), "\n")
```

TCPI: 1.429

```
R> cat("VAC:  $", format(round(vac(bac, eac_typ)), big.mark = ","), "\n")
```

VAC: \$ -225,000

The To-Complete Performance Index (TCPI) reports the cost efficiency required on all remaining work to finish within the original budget; values substantially above 1.0 signal an unrealistic recovery target.

4. Bayesian Risk Methods

Building on the causal structure introduced in Govan (2014) and Govan and Damnjanovic (2016), the Bayesian module models risk event probabilities from root causes and supports formal updating as project observations become available (Gelman, Carlin, Stern, Dunson, Vehtari, and Rubin 2013; Pearl 2009).

4.1. Prior Risk Probability

`risk_prob()` computes the probability of a risk event R from multiple independent root causes. Each cause contributes a marginal probability by the law of total probability, and the causes are combined with a noisy-OR—the probability that at least one cause triggers R —so the result always lies in $[0, 1]$:

$$P(R) = 1 - \prod_i \left[1 - (P(R | C_i) P(C_i) + P(R | \bar{C}_i) P(\bar{C}_i)) \right]$$

For the `building_project` schedule-delay risk with two root causes (adverse weather and late design changes):

```
R> prior <- risk_prob(bp$cause_probs, bp$risks_given_causes,
+                    bp$risks_given_not_causes)
R> cat("Prior P(delay):", round(prior, 3), "\n")
```

Prior P(delay): 0.578

4.2. Bayesian Updating

`risk_post_prob()` updates the risk probability after observing cause status on site. An observed cause is fixed to its conditional probability; unobserved causes are passed as `NA` and retain their prior (marginal) contribution.

```
R> # Weather observed to have occurred; design changes not yet assessed
R> post <- risk_post_prob(bp$cause_probs, bp$risks_given_causes,
+                         bp$risks_given_not_causes, bp$observed_causes)
R> cat("Posterior P(delay):", round(post, 3), "\n")
```

```
Posterior P(delay): 0.797
```

```
R> cat("Change:           ", round(post - prior, 3), "\n")
```

```
Change:           0.219
```

Confirming that adverse weather has occurred raises the delay probability relative to the prior. The posterior informs contingency updates mid-project.

4.3. Cost Risk Distributions

`cost_pdf()` and `cost_post_pdf()` sample prior and posterior cost distributions by simulating which risks materialize and summing their cost impacts.

```
R> risk_probs          <- c(0.3, 0.5, 0.2)
R> means_given_risks  <- c(10000, 15000, 5000)
R> sds_given_risks    <- c(2000, 1000, 1000)
R> base_cost          <- 2000
R>
R> prior_costs <- cost_pdf(5000, risk_probs, means_given_risks,
+                         sds_given_risks, base_cost)
R> post_costs  <- cost_post_pdf(5000,
+   observed_risks = c(1, NA, 0), # Risk 1 present, Risk 2 unobserved, Risk 3 absent
+   means_given_risks, sds_given_risks, base_cost
+ )
R>
R> plot(density(prior_costs), col = "steelblue", lwd = 2,
+      main = "Cost Risk: Prior vs. Posterior",
+      xlab = "Cost ($)", xlim = range(c(prior_costs, post_costs)))
R> lines(density(post_costs), col = "tomato", lwd = 2)
R> legend("topright", legend = c("Prior", "Posterior"),
+      col = c("steelblue", "tomato"), lwd = 2, bty = "n")
```

4.4. Correlation Matrix Construction

`cor_matrix()` constructs a valid positive-definite correlation matrix from a vector of upper-triangle entries, for use as input to `mcs()`.

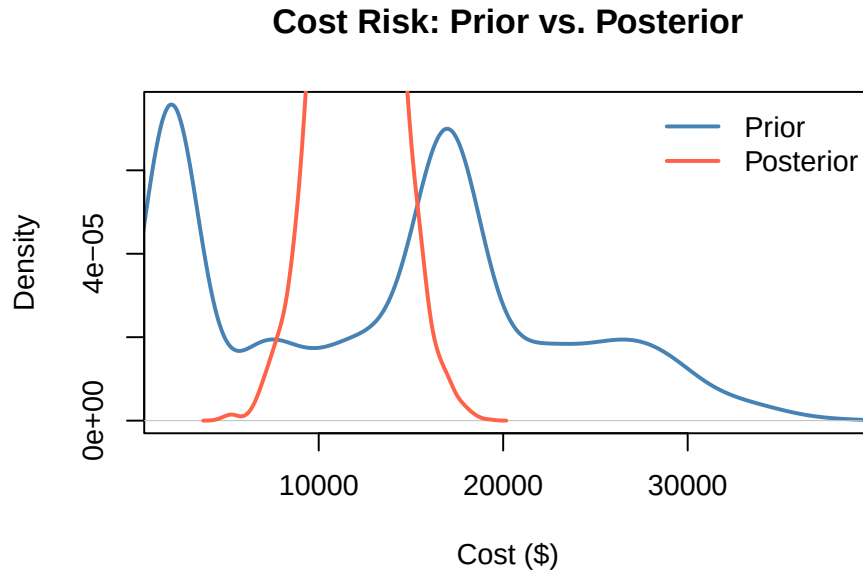


Figure 3: Prior and posterior cost distributions. Observing Cause 1 shifts the distribution rightward, increasing the expected cost impact.

```
R> dists <- list(
+   normal = function(n) rnorm(n, mean = 10, sd = 2),
+   triang = function(n) mc2d::rpert(n, min = 5, mode = 10, max = 15),
+   uniform = function(n) runif(n, min = 8, max = 12)
+ )
R> C_emp <- cor_matrix(num_samples = 1000, num_vars = 3, dists = dists)
R> print(round(C_emp, 3))

      [,1] [,2] [,3]
[1,] 1.000 0.024 0.000
[2,] 0.024 1.000 0.003
[3,] 0.000 0.003 1.000
```

5. Learning Curves

Sigmoidal (S-curve) models capture the characteristic pattern of project progress: slow start, rapid acceleration, and plateau approaching 100% completion. **PRA** provides three model types—Logistic, Pearl, and Gompertz—fitted via the Levenberg–Marquardt nonlinear least-squares algorithm in **minpack.lm** (Elzhov *et al.* 2016).

```
R> progress <- data.frame(
+   time      = 1:9,
+   completion = c(5, 15, 40, 60, 70, 75, 80, 85, 90)
+ )
```

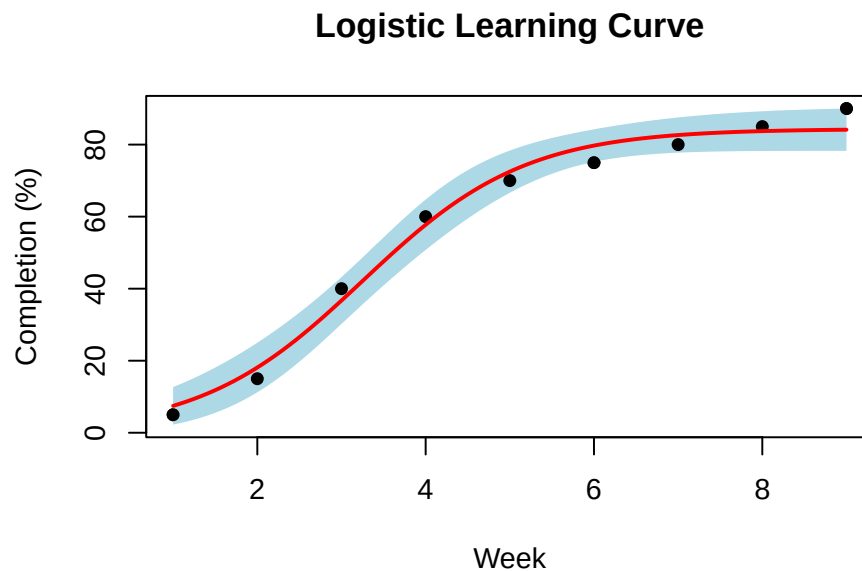


Figure 4: Logistic learning curve with 95% confidence band. The shaded region widens as extrapolation extends beyond the observed data range.

```
R> fit_log <- fit_sigmodal(progress, "time", "completion", "logistic")
R> fit_gomp <- fit_sigmodal(progress, "time", "completion", "gompertz")
R>
R> cat("Logistic RSE: ", round(summary(fit_log)$sigma, 2), "\n")
```

```
Logistic RSE: 4.15
```

```
R> cat("Gompertz RSE: ", round(summary(fit_gomp)$sigma, 2), "\n")
```

```
Gompertz RSE: 2.89
```

```
R> plot_sigmodal(fit_log, progress, "time", "completion", "logistic",
+               conf_level = 0.95,
+               main = "Logistic Learning Curve",
+               xlab = "Week", ylab = "Completion (%)")
```

`predict_sigmodal()` generates forecasts with confidence bounds at arbitrary future time points:

```
R> preds <- predict_sigmodal(fit_log, seq(10, 12, by = 0.5),
+                           "logistic", conf_level = 0.95)
R> print(round(preds, 1))
```

```
      x pred lwr upr
1 10.0 84.3 78.2 90.3
```

```

2 10.5 84.3 78.2 90.4
3 11.0 84.3 78.2 90.4
4 11.5 84.3 78.2 90.5
5 12.0 84.3 78.2 90.5

```

6. Dependency Structure Analysis

Building on Govan and Damnjanovic (2016, 2020) and the foundational DSM literature ([Steward 1981](#); [Browning 2001](#)), the DSM module quantifies structural coupling between project tasks through shared resources and risks.

6.1. Design Structure Matrices

The **Parent DSM** is derived from the Resource–Task matrix $\mathbf{S}$ (rows = resources, columns = tasks) as $\mathbf{P} = \mathbf{S}^\top \mathbf{S}$. Off-diagonal entry P_{jk} counts the number of resources shared between tasks j and k —the structural pathway through which disruptions propagate ([Govan and Damnjanovic 2016](#)).

```

R> S <- matrix(c(
+   1, 0, 1, 0,
+   1, 1, 0, 0,
+   0, 1, 0, 1,
+   0, 0, 1, 1,
+   0, 1, 1, 0
+ ), nrow = 4, ncol = 5)
R> rownames(S) <- paste0("R", 1:4)
R> colnames(S) <- paste0("T", 1:5)

R> p <- parent_dsm(S)
R> plot(p)

```

The **Grandparent DSM** adds a risk layer via the Risk–Resource matrix $\mathbf{R}$, tracing dependencies from risks through resources to tasks. Off-diagonal entry G_{jk} counts the number of risks shared between task j and task k through the resource chain—a direct operationalization of the structural network measures developed in Govan and Damnjanovic (2020).

```

R> R_mat <- matrix(c(
+   1, 0, 1,
+   1, 1, 0,
+   0, 1, 0,
+   0, 0, 1
+ ), nrow = 3, ncol = 4)
R> rownames(R_mat) <- paste0("Risk", 1:3)
R> colnames(R_mat) <- paste0("R", 1:4)

R> g <- grandparent_dsm(S, R_mat)
R> plot(g)

```

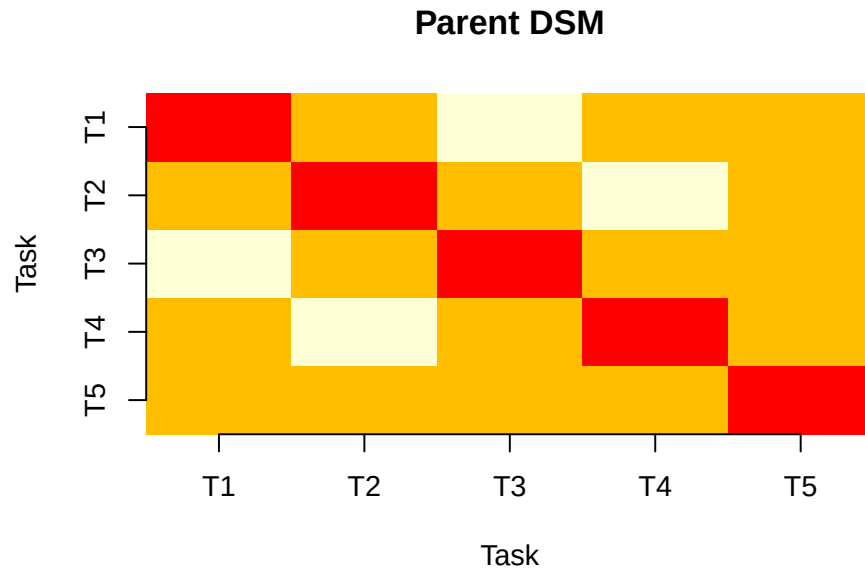


Figure 5: Parent DSM heatmap. Darker cells indicate more shared resources and stronger coupling between task pairs.

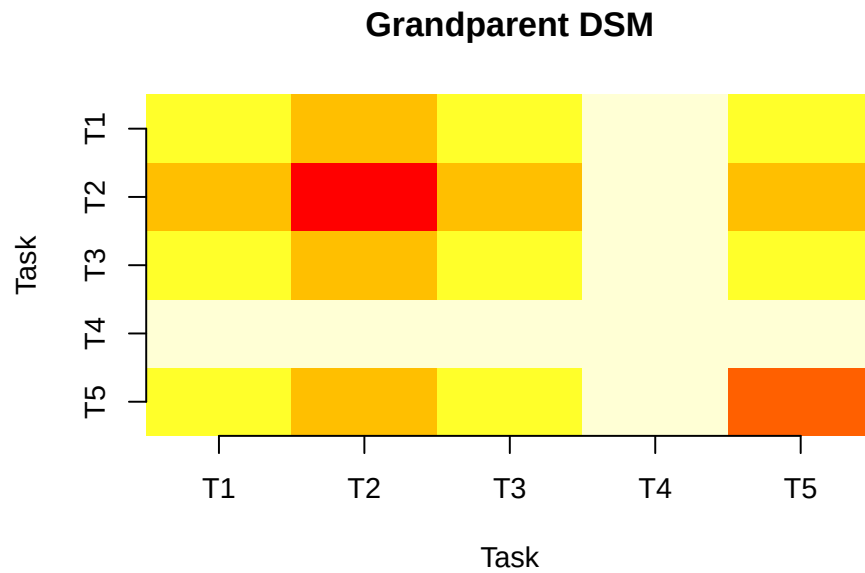


Figure 6: Grandparent DSM heatmap. Shows risk-based coupling: task pairs that share risk exposure through common resources tend to fail together.

Task pairs with high off-diagonal values in the Grandparent DSM represent concentrated risk exposure; these pairs are prime candidates for contingency buffering or resource decoupling.

7. Probabilistic Networks

Experimental. The probabilistic network module extends DSM structural analysis by attaching probability distributions to each node, enabling forward simulation of cost uncertainty and Bayesian conditioning on observed evidence. A probabilistic network is a directed acyclic graph in which nodes represent project variables (risks, resources, tasks, and the project total) and edges encode conditional dependencies (Pearl 2009; Govan 2014). Where a DSM counts shared resources between task pairs, a probabilistic network propagates distributions through the causal chain from risks to total project cost.

7.1. Building a Network

`prob_net()` constructs the network from a node data frame, a links data frame, and a named list of distributions. Distributions may be `normal`, `lognormal`, `uniform`, `discrete`, or `conditional` (switching between two sub-distributions depending on a discrete parent node); `aggregate` nodes sum their parents.

```
R> nodes <- data.frame(
+   id = c("A","B","C","D","E","F","G","H","I"),
+   label = c("Risk-1","Risk-2","Resource-1","Resource-2","Resource-3",
+             "Task-1","Task-2","Task-3","Project"),
+   stringsAsFactors = FALSE
+ )
R> links <- data.frame(
+   source = c("A","B","C","D","E","F","G","H"),
+   target = c("C","D","F","G","H","I","I","I"),
+   stringsAsFactors = FALSE
+ )
R> distributions <- list(
+   A = list(type = "discrete", values = c(1, 0), probs = c(0.70, 0.30)),
+   B = list(type = "discrete", values = c(1, 0), probs = c(0.60, 0.40)),
+   C = list(type = "conditional", condition = "A",
+     true_dist = list(type = "normal", mean = 30000, sd = 8000),
+     false_dist = list(type = "normal", mean = 15000, sd = 3000)),
+   D = list(type = "conditional", condition = "B",
+     true_dist = list(type = "normal", mean = 80000, sd = 20000),
+     false_dist = list(type = "normal", mean = 50000, sd = 10000)),
+   E = list(type = "normal", mean = 20000, sd = 4000),
+   F = list(type = "aggregate", nodes = "C"),
+   G = list(type = "aggregate", nodes = "D"),
+   H = list(type = "aggregate", nodes = "E"),
+   I = list(type = "aggregate", nodes = c("F","G","H"))
+ )
R> net <- prob_net(nodes, links, distributions = distributions)
```

7.2. Forward Simulation

`prob_net_sim()` draws samples from every node in topological order and returns a data frame with one column per node.

```
R> sim_net <- prob_net_sim(net, num_samples = 10000)
R> cat("Mean project cost: $", format(round(mean(sim_net$I)), big.mark = ","), "\n")
```

```
Mean project cost: $ 113,572
```

```
R> cat("P80 project cost: $", format(round(quantile(sim_net$I, 0.80)),
+   big.mark = ","), "\n")
```

```
P80 project cost: $ 135,657
```

7.3. Learning: Conditioning on Evidence

`prob_net_learn()` clamps observed nodes to fixed values and re-simulates the downstream distribution—an observational update in the sense of [Pearl \(2009\)](#). The following call fixes Risk-2 (node B) to absent (value 0), ruling out Technical Complexity as a concern:

```
R> learn_net <- prob_net_learn(net, observations = list(B = 0),
+   num_samples = 10000)
R>
R> plot(density(sim_net$I), col = "steelblue", lwd = 2,
+   main = "Total Project Cost: Prior vs. Posterior",
+   xlab = "Cost ($)")
R> lines(density(learn_net$I), col = "tomato", lwd = 2)
R> legend("topright",
+   legend = c("Prior (Risk-2 uncertain)", "Posterior (Risk-2 = 0)"),
+   col = c("steelblue", "tomato"), lwd = 2, bty = "n")
```

`prob_net_update()` performs structural surgery on the network by adding or removing edges and replacing node distributions, corresponding to the *do*-operator in Pearl’s causal calculus ([Pearl 2009](#))—distinct from `prob_net_learn()`, which conditions without altering the graph. The probabilistic network API is marked experimental and may evolve in future releases.

8. Model Context Protocol Integration

Version 0.5.0 of **PRA** exposes all analytical functions as tools callable by large-language-model (LLM) agents through the Model Context Protocol (MCP) ([Anthropic 2024](#)). `pra_mcp_server()` starts a server that advertises each tool—Monte Carlo simulation, EVM, Bayesian risk, learning curves, DSM, and the rest—so that Claude, Claude Code, and other MCP-compatible clients can invoke them as native tools. The server is registered in Claude Code with a single shell command:

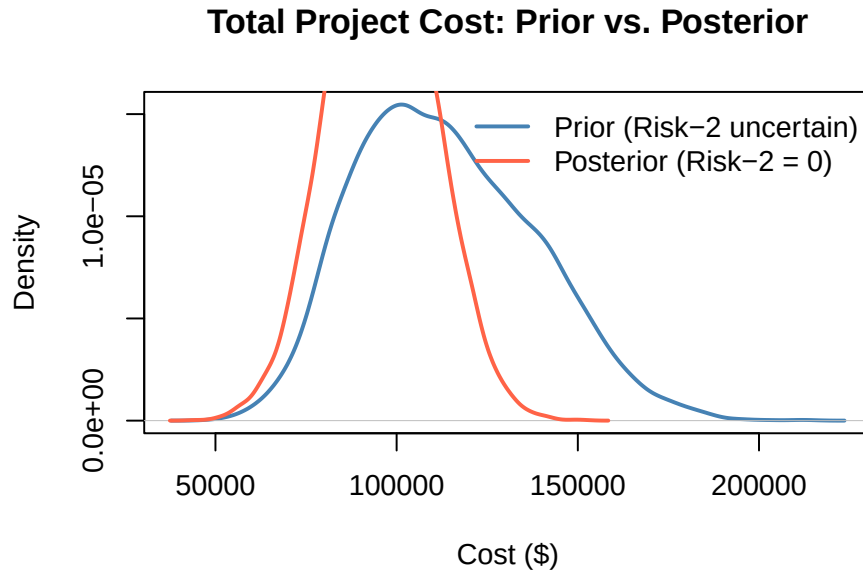


Figure 7: Total project cost before and after observing that Risk-2 (Technical Complexity) did not occur. Conditioning on Risk-2 = 0 eliminates the heavy upper tail and shifts the distribution leftward by roughly \$30,000 in expectation.

```
R> claude mcp add -s project pra -- Rscript -e "PRA::pra_mcp_server()"
```

For Claude Desktop, add the following block to the MCP configuration file:

```
{
  "mcpServers": {
    "pra": { "command": "Rscript",
              "args": ["-e", "PRA::pra_mcp_server()"] }
  }
}
```

The tool definitions are built once with `ellmer::tool()` (Wickham *et al.* 2025) and served by **mcptools**. Each tool validates its JSON inputs and returns a plain-text summary for the LLM together with, where applicable, a rich HTML result with inline plots for clients that render it.

To help an agent decide *when* and *how* to call each tool, **PRA** bundles a set of Agent Skills under `system.file("skills", package = "PRA")`. Each skill is a `SKILL.md` file with structured guidance—when to use the method, which tool to call, the JSON argument format, and how to interpret the result—covering Monte Carlo simulation, sensitivity and contingency, EVM, Bayesian risk, learning curves, and dependency structure analysis, plus a routing overview. Pairing the skills with the MCP server lets an agent carry out a complete project risk workflow (for example, `mcs_tool` followed by `contingency_tool` and `sensitivity_tool`) from a natural-language request, without embedding an LLM in the package itself.

The MCP layer is strictly optional: every analytical function operates without any optional dependency. Only `pra_mcp_server()` requires **ellmer**, **mcptools**, and **jsonlite**.

9. Illustrative Case Study

This section demonstrates an integrated, end-to-end project risk analysis for the `building_project` dataset introduced in Section~2, progressing through uncertainty quantification, performance monitoring, Bayesian risk updating, and dependency structure analysis in a single R session.

9.1. Step 1: Characterize Task Uncertainty

```
R> bp <- building_project
R> set.seed(42)
R> sim <- mcs(10000, bp$task_distributions)
R> res <- contingency(sim, phigh = 0.80, pbase = 0.50)
R> cat("P50 schedule:", round(quantile(sim$total_distribution, 0.50), 1), "weeks\n")
```

P50 schedule: 71.6 weeks

```
R> cat("P80 schedule:", round(quantile(sim$total_distribution, 0.80), 1), "weeks\n")
```

P80 schedule: 75.6 weeks

```
R> cat("Contingency (P80-P50):", round(res, 2), "weeks\n")
```

Contingency (P80-P50): 3.98 weeks

9.2. Step 2: Monitor Earned Value at Mid-Project

```
R> pv_cs <- pv(bp$bac, bp$schedule, bp$time_period)
R> ev_cs <- ev(bp$bac, bp$actual_per_complete)
R> ac_cs <- ac(bp$actual_costs, bp$time_period, cumulative = FALSE)
R>
R> cat("SPI:", round(spi(ev_cs, pv_cs), 3),
+      " CPI:", round(cpi(ev_cs, ac_cs), 3), "\n")
```

SPI: 0.889 CPI: 0.986

```
R> cat("EAC (typical): $", format(round(eac(bp$bac,
+      method = "typical", cpi = cpi(ev_cs, ac_cs))), big.mark = ","), "\n")
```

EAC (typical): \$ 8,625,000

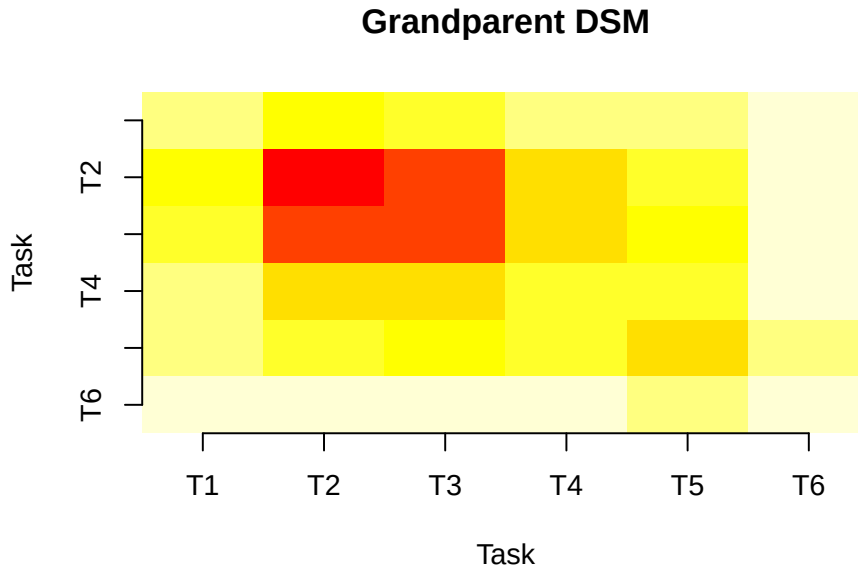


Figure 8: Grandparent DSM for the case-study project, showing which task pairs share the most risk exposure through common resources.

9.3. Step 3: Update Risk Probability After Site Inspection

```
R> prior_r <- risk_prob(bp$cause_probs, bp$risks_given_causes,
+                       bp$risks_given_not_causes)
R> post_r  <- risk_post_prob(bp$cause_probs, bp$risks_given_causes,
+                           bp$risks_given_not_causes, bp$observed_causes)
R> cat("Prior P(delay):", round(prior_r, 3),
+      "-> Posterior P(delay):", round(post_r, 3), "\n")
```

Prior P(delay): 0.578 -> Posterior P(delay): 0.797

9.4. Step 4: Identify Coupled Tasks via DSM

```
R> g_cs <- grandparent_dsm(bp$resource_task, bp$risk_resource)
R> plot(g_cs)
```

The Grandparent DSM reveals which task pairs share risk exposure through common resources, informing targeted contingency allocation and resource decoupling decisions.

10. Summary

PRA is a CRAN-available R package that provides a unified, open-source implementation of quantitative project risk analysis methods. It operationalizes the causal network and structural network frameworks developed in Govan (2014) and Govan and Damnjanovic (2016, 2020) through ten integrated analytical modules: SMM, MCS, contingency analysis, sensitivity analysis, EVM (11 metrics), Bayesian risk inference and updating, cost risk distributions, sigmoidal learning curves, correlation matrix construction, and design structure matrices. An experimental eleventh module provides probabilistic networks with forward simulation and Bayesian conditioning.

Version 0.5.0 exposes every analytical function as a tool over the Model Context Protocol, complemented by a bundled set of Agent Skills, so that LLM agents can drive the package without any embedded model. All exported functions adhere to rOpenSci SRR standards, with a test suite verifying correctness, edge-case handling, and numerical stability across three operating systems. The package is licensed under MIT and the source is available at <https://github.com/paulgovan/PRA>; documentation and a companion online book are at <https://paulgovan.github.io/PRA/>.

10.1. Limitations and Future Work

Several limitations apply. First, **PRA** focuses on quantitative uncertainty propagation; it does not model schedule logic (critical path, activity dependencies, or resource leveling), which remain the domain of dedicated scheduling engines. Second, `mcs()` induces task correlation by applying a Cholesky factor to the sampled values, which reproduces the target rank correlation but distorts the marginal means; the examples in this article therefore treat tasks as independent for Monte Carlo simulation and use `smm()`, which handles correlation analytically, when correlation matters. A copula-based implementation that preserves the marginals is planned. Third, the probabilistic network module is experimental: the API may change and no convergence diagnostics are provided for the Monte Carlo estimators. Fourth, the package currently accepts data as R vectors and lists; direct import from project management file formats (Microsoft Project, Primavera P6, CSV exports from Jira or Asana) is not supported and is planned for a future release.

Computational details

The results in this article were produced using R (R Core Team 2024) with the **PRA** package (version 0.5.0) and its hard dependencies **mc2d** (Pouillot and Delignette-Muller 2010) and **minpack.lm** (Elzhov *et al.* 2016). The optional Model Context Protocol server additionally uses **ellmer** (Wickham *et al.* 2025), **mcptools**, and **jsonlite**. R itself and all packages used are available from the Comprehensive R Archive Network (CRAN) at <https://CRAN.R-project.org/>. A random seed of 42 is set at the start of the session so that the simulation output is reproducible.

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Affiliation: